

Kaifeng “Kevin” Jia

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EDUCATION

Master of Engineering in Electrical and Computer Engineering,

Cornell University

Aug. 2025 – Dec. 2026 (expected)

- Core courses: Fast Robots, Autonomous Mobile Robots, Embedded OS, Natural Language Processing.

Bachelor of Engineering in Optoelectronic Information Science and Engineering,

Hefei University of Technology

Sept. 2021 – May 2025

- Concentration: Signals and Systems, Digital Image Processing, Sensor Principles and Technology.

PROFESSIONAL SKILLS

Machine Learning & RL: Deep Learning, Reinforcement Learning, Multi-Agent RL, PPO

Programming Languages: Python, C/C++, Swift

Robotics & Perception: SLAM, EKF, Bayes filtering, PID control, sensor fusion (IMU/ToF), path/motion planning, YOLO

Simulation & Tools: Isaac Lab / Isaac Sim, PyTorch, Git, Linux, MATLAB

RESEARCH EXPERIENCE

Multi-Agent Reinforcement Learning for Cooperative Loco-Manipulation

Carnegie Mellon University

Advised by Dr. Hao Zhang

Jun. 2026 – Present

- Developing an **object-aware multi-agent RL** framework in which multiple humanoid robots (Unitree G1) learn to cooperatively manipulate full-scale objects in large-scale simulation (**Isaac Sim**), autonomously emerging diverse cooperative contact strategies.
- Initialized manipulation skills from **OmniRetarget**-based demonstration retargeting; designed multi-agent RL that generalizes across object mass, size, and geometry and produces role specialization rather than replicated single-agent policies.

Depth Foundation Model Adaptation via Nanophotonic Wavefront Prompting

New York University

Advised by Bingrui Li (PhD student, NYU)

May. 2026 – Present

- Building on “Physically Grounded Monocular Depth via Nanophotonic Wavefront Prompting” (Li et al., **ECCV 2026**), exploring a **new architecture** and adaptation module to align a pretrained **depth foundation model** with metalens polarization optical signals for single-shot monocular **metric depth estimation**.

PROJECTS

Fast Robots: Autonomous RC Car Platform,

Cornell University, Jan. 2026 – May. 2026

- Built full **localization and planning** pipeline — ToF-based 360° environment mapping, **Bayes filter** grid localization on a real robot, and autonomous navigation through 9 waypoints via orientation PID and an onboard state machine.
- Designed a closed-loop control system with **PID controllers** for distance hold and orientation, augmented by a **Kalman filter** with system-identified drag and momentum parameters; implemented an autonomous drift stunt via a ToF-triggered state machine.
- Developed full hardware integration on an **Artemis microcontroller** — wiring and configuring **VL53L1X ToF**, **ICM-20948 IMU**, motor drivers, and a **BLE** stack for real-time sensor streaming and command execution.

Indoor AR Navigation System — Olin Library, Cornell University,

Sept. 2025 – Present

- Built an iOS AR indoor navigation system using **ARKit visual-inertial SLAM** to estimate device pose and render real-time navigation guidance on the live camera feed.
- Implemented **map-based relocalization** via pre-recorded visual feature maps, enabling automatic pose recovery upon re-entry without manual initialization.
- Constructed a metric-scale **walkable graph** from floor plans and applied **Dijkstra’s algorithm** for path planning; designed a turn-detection module with angular thresholding and 3D AR directional cue rendering.

Smart Desk Assistant: Multi-modal Raspberry Pi System

Cornell University, Aug. 2025 – Dec. 2025

- Built a **multi-modal desktop assistant** on Raspberry Pi integrating a touch UI (**PyGame/PiTFT**), voice interaction (**Vosk ASR + LLM API + TTS**), and **DNN-based face tracking** driven by a PID servo controller with a state machine for wake-word detection, speech-to-text, and LLM response generation.